

THE DETERMINATION OF TINNITUS LOUDNESS CONSIDERING THE EFFECTS OF RECRUITMENT

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We have attempted to determine a meaningful measure of tinnitus loudness that considers loudness recruitment. Sixteen subjects with sensorineural tinnitus adjusted the level of a pure tone so that it was (a) at threshold, (b) equal in loudness to the tinnitus, and (c) uncomfortably loud. The pure-tone frequency was (a) at the frequency of the tinnitus pitch, and (b) at the octave frequency between 500 and 4000 Hz having the pure-tone threshold closest to 0 dB HL. In addition, the subjects adjusted the level of a broadband noise so that it (a) was at threshold, (b) just masked the tinnitus, and (c) was uncomfortably loud. These measurements were performed both in the ear ipsilateral and in the ear contralateral to the tinnitus. Formulae based on abnormal loudness functions and uncomfortable loudness levels are presented that convert equal-loudness matches in dB sensation level (SL) into loudness in sones. The level of broadband noise (in dB SL) required to mask tinnitus was about the same in the ipsilateral and in the contralateral ear for most of our subjects, regardless of the ear in which the tinnitus was localized. This noise level (in dB SL) required to mask the tinnitus correlated well with the level (in dB SL) of a tone at the most normal frequency judged equal in loudness to the tinnitus.

Fowler (1940) suggested that tinnitus loudness could be determined by matching the tinnitus to the loudness of a pure tone presented to the ear contralateral to the tinnitus. The threshold of this tone is then obtained, and the "loudness" is interpreted as the dB sensation level (SL) of the pure tone. Although many patients complain of the extreme annoyance of their tinnitus (Institute of Hearing Research, 1981; Tyler & Baker, in press), these loudness balance procedures usually produce loudness estimates between 0-20 dB SL. Some typical loudness balancing results from a number of studies are shown in

Table 1. About 88% of the subjects in these studies matched their tinnitus loudness to a tone of less than 20 dB SL. In addition to those listed, Graham and Newby (1962) indicated that over half of 73 subjects matched their tinnitus to a tone of 5 dB SL or less. However, as Stevens and Davis mentioned in 1938, "the decibel, contrary to what is sometimes assumed, is not a unit of loudness" (p. 110). Stevens (1955) later developed the phon and sone scales to quantify the perception of loudness (for a review, see Scharf, 1978). We believe these scales can be adapted to help understand the loudness of

TABLE 1. Results of loudness balancing procedures from studies in which a pure tone or narrow-band noise is equated in loudness to the tinnitus. The categories along the top of the table show the range of pure-tone levels (in dB SL) equated in loudness to the tinnitus. Each entry represents the number of subjects in each dB SL category for a particular study. The test ear is designated SF (sound field), C (contralateral), and I (ipsilateral). The ? indicates the test ear was not clearly defined.

Study	Test ear	Level of pure tone judged equally loud dB SL				
		0-10	11-20	21-30	31-40	41-50
Mortimer, Burr, Wright, & McGarry (1940)	SF	6	5	1	0	0
Reed (1960)	C	62	16	7	3	2
Donaldson (1978)	C?	7	9	4	0	0
Bailey (1979)	C?	15	1	0	0	0
Roeser & Price (1980)	I?	64	9	10	0	0
Shailer, Tyler, & Coles (1981)	C	9	1	1	0	0
Present study	I	14	1	0	1	0
Total		177	42	23	4	2
Percentage		71%	17%	9%	2%	1%

Note. In all cases the frequency of the pure tone or narrow-band noise was equal to the pitch-match frequency of the tinnitus.

TABLE 2. Biographic and questionnaire data for 16 subjects with tinnitus. The tinnitus location is indicated by R (right), L (left), H (head), or B (both ears). The letter in brackets defines the ear ipsilateral to the tinnitus (see text). Subjects estimated the loudness (1-very soft, 6-very loud) and pitch (1-very low, 6-very high) of their tinnitus.

Subject	Age (Yrs)	Sex	Tinnitus description	Tinnitus location	Suspected etiology	Tinnitus duration (Yrs)	Loudness estimation (1-6)	Pitch estimation (1-6)	Pitch-match frequency (Hz)
1	29	F	Rush/throb	R	Vascular	6	4	4	441
2	54	M	Whistling	B (L)	Viral	40	3	5	7189
3	55	M	Escaping steam	B/H (L)	Menieres	6	6	4	5114
4	65	F	Rushing	R	Head injury	2	5	5	380
5	68	F	Rush/throb	L/H	Unknown	6	4	4	6150
6	80	M	Buzz/steam	L	Unknown	4	4	4	5873
7	72	F	Steam	H (L)	Unknown	18	4	4	4856
8	61	F	Ringling	L	Unknown	40	6	6	7490
9	62	F	Ring/throb	L	Unknown	7	4	5	2104
10	58	M	Ringling	H (R)	Unknown	10	4	6	9281
11	63	F	Ringling	L	Head injury	14	2	3	1226
12	79	M	Ringling	B (L)	Unknown	30	4	4	8128
13	74	M	Whistling	R	Viral	30	6	6	2832
14	59	F	Ringling	H (L)	Unknown	7	4	4	667
15	56	M	Whistle/ring	B/H (L)	Ototoxic drug	7	5	6	11470
16	58	M	Whistling	B (L)	Noise induced	10	6	6	6532

tinnitus. Loudness recruitment should be considered also, since the majority of people with tinnitus also have some hearing loss.

Describing tinnitus loudness with the sone scale should better represent the true tinnitus loudness. This measurement could (a) have diagnostic significance, (b) provide a quantitative measure to monitor tinnitus deterioration or improvement during therapy, (c) provide part of a comprehensive tinnitus classification scheme, and (d) provide a more meaningful psychoacoustical account of some of the severe discomfort and annoyance caused by tinnitus.

In the present study we used loudness balancing procedures to match the loudness of a pure tone to the tinnitus loudness. In the discussion we first provide formulae which convert dB SL equal-loudness measurements into tinnitus loudness in sones and then apply these formulae to our data. In addition we measured the level of broadband noise required to mask tinnitus completely, to determine whether this measurement has any relationship to tinnitus loudness.

METHOD

Subjects

All 16 subjects were members of the British Tinnitus Association resident in the Nottingham area. They were paid one pound per hour and travelling expenses. Thirteen subjects had no previous experience with tinnitus research experiments. Three subjects had participated in a pitch-matching experiment performed three months earlier. Relevant biographical details for all subjects are given in Table 2, and audiometric data are given in Table 3.

TABLE 3. Audiograms for the 16 patients with tinnitus. Pure-tone air-conduction thresholds are shown in dB HL (re ISO-389, 1975).

Subject	Left ear				Right ear			
	.5	1.0	2.0	4.0	.5	1.0	2.0	4.0
		(kHz)				(kHz)		
1	20	5	10	-5	45	30	10	5
2	15	10	15	15	15	5	10	10
3	10	10	25	65	10	10	30	60
4	20	30	45	40	60	60	55	45
5	20	15	25	20	25	15	30	30
6	25	20	30	35	25	25	30	40
7	15	35	65	70	20	30	80	75
8	20	20	20	25	10	10	20	30
9	15	35	55	40	10	10	20	10
10	5	5	5	30	10	5	5	30
11	35	45	35	40	25	35	40	50
12	30	30	55	70	70	65	90	100
13	35	50	45	60	75	75	70	90
14	20	15	30	40	15	15	35	40
15	10	10	15	25	10	10	10	15
16	35	50	65	60	35	50	60	60

Note. Thresholds for bone-conduction tones were within 5 dB of the air-conduction thresholds for each frequency.

Procedures

The ipsilateral ear was defined as the ear or side of the head judged by the subject to contain the tinnitus or the loudest tinnitus in bilateral cases. If a subject with bilateral tinnitus judged it to be of equal loudness in both ears, then the ipsilateral ear was the one in which the tinnitus was judged to be most steady in pitch.

Three pitch-match trials were performed in the ipsilateral ear using the method of adjustment. The subject was instructed to adjust the frequency dial until the

pitch of a pulsed 1-sec pure tone matched the pitch of the tinnitus. The details of the procedure and stimulus parameters are identical to those used by Tyler and Conrad-Armes (Note 1). After each trial, tones were presented at the pitch-match frequency and one octave above and one octave below it to test for octave confusions. The subject was asked to indicate which of these three tones was nearest to the tinnitus pitch. This value was accepted as the pitch match. The median of the three pitch matches was recorded as the tinnitus pitch.

We measured (a) threshold, (b) the level judged to be equal in loudness to the tinnitus, and (c) the uncomfortable loudness level (ULL) for a pure tone at two values of the frequency of the pure tone: the tinnitus pitch-match frequency, and the frequency of the most normal threshold at .5, 1.0, 2.0, or 4.0 kHz. The instructions were:

(for threshold)	Set the level of the tone so that you can only just hear it pulsing.
(for equal loudness)	Set the level of the tone so that it is equal in loudness to your tinnitus.
(for the ULL)	Set the level of the tone so that the loudness is just at a level which is uncomfortable for you to listen to for more than a few seconds.

Each condition was repeated five times and an average value obtained.

Using broadband noise, we also measured the threshold, the masked threshold, and the ULL. The instructions for the masked threshold were: "Set the level of the noise so that it is just loud enough to make you unable to hear your tinnitus." Each broadband noise condition was repeated three times and an average value obtained.

Subjects were instructed to turn the attenuator dial through a wide range of intensities and gradually center

in on the appropriate level. A second attenuator was controlled by the tester and set to a different value prior to each condition, so that the subjects could not use the attenuator dial markings in their judgments. All conditions were presented in random order.

Apparatus

A signal/noise generator (Wavetek 132) produced the pure-tone signal or the broadband noise. The stimulus was fed to a gate (Coulbourn S84-04), a Marconi attenuator (TF 21-62), and a manual attenuator covering a 120-dB range (30 dB of this range in 1-dB steps). The output went to a Marantz Amplifier (1072) and a single Telephonics TDH-49 earphone encased in an MX-41/AR cushion. Testing was performed in a sound-proof booth.

The stimuli rise-fall times were 40 msec (from 10 to 90% levels on the signal envelope), and the duration was 1 sec (between 50% levels on the stimulus envelope). The stimuli were pulsed with an off-time of 1 sec. The harmonics of the pure tones were at least 55 dB below the level of the fundamental. The electrical spectrum of the broadband noise was flat ± 1 dB between 0–10,000 Hz at the input to the earphone. The frequency response of the earphone showed a slight peak around 3500 Hz and dropped off rapidly beyond 7000 Hz (as measured in a Bruel & Kjaer 4153 artificial ear).

RESULTS

Loudness Matching

Table 4 displays the individual results for the ipsilateral measurements. The left side of the table shows the results obtained at the frequency of the tinnitus pitch, and the right side shows the results obtained at the frequency of the most normal pure-tone threshold. The

TABLE 4. Loudness matching data (in dB SL) measured in the ear ipsilateral to the tinnitus. (Thresh = pure-tone threshold; ULL = Uncomfortable Loudness Level.)

Subject	Ear ipsilateral to tinnitus					
	At frequency of tinnitus pitch			At frequency of most normal threshold		
	Thresh	Equal loudness	ULL	Thresh	Equal loudness	ULL
1	20.6	43.0	66.6	-4.2	37.8	65.8
2	42.4	.6	58.6	13.8	1.4	89.0
3	77.6	8.4	26.8	38.0	32.0	62.4
4	53.0	9.6	47.6	42.0	15.2	50.0
5	61.2	7.0	38.8	27.0	23.8	60.2
6	84.0	1.8	8.8	37.0	11.6	36.2
7	59.2	4.4	9.6	18.6	12.0	32.2
8	78.2	12.4	17.2	46.2	23.6	42.2
9	34.6	7.4	13.8	4.2	29.4	39.4
10	79.4	5.2	20.4	16.0	62.0	89.8
11	37.4	7.6	40.2	44.4	7.6	30.0
12	105.0	5.2	9.0	39.6	-.6	30.6
13	84.2	.4	5.2	81.6	2.6	8.4
14	35.0	33.6	46.6	26.4	12.4	23.8
15	79.8	1.2	21.2	23.4	44.8	85.2
16	83.8	1.0	6.2	39.4	19.2	52.0

TABLE 5. Loudness matching data (in dB SL) from 16 subjects with tinnitus measured in the ear contralateral to the tinnitus. (Thresh = pure-tone threshold; ULL = Uncomfortable Loudness Level.)

Subject	Ear contralateral to tinnitus					
	At frequency of tinnitus pitch			At frequency of most normal threshold		
	Thresh	Equal loudness	ULL	Thresh	Equal loudness	ULL
1	55.8	8.4	27.6	1.4	37.2	59.8
2	29.8	11.0	71.2	15.6	1.6	84.0
3	78.4	6.2	25.2	35.8	24.2	61.8
4	74.6	10.6	31.0	57.2	19.8	29.8
5	62.6	15.4	39.8	25.4	28.6	60.2
6	61.6	1.4	12.2	38.0	12.4	35.4
7	70.8	6.0	9.6	14.2	17.8	31.6
8	76.2	5.8	18.8	26.8	28.4	53.8
9	3.4	38.8	46.2	-2.6	33.2	54.0
10	60.0	6.4	25.2	17.4	53.0	84.6
11	41.8	1.4	31.0	39.8	14.0	32.2
12 ^a	-	-	-	81.0	10.0	26.6
13	55.8	2.8	15.0	48.6	-.2	6.4
14	32.8	16.2	37.2	24.2	18.8	31.2
15	92.2	1.8	8.8	21.6	36.8	83.8
16	70.8	7.6	14.2	45.4	15.2	43.4

^aSubject 12 had a profound pure-tone threshold loss at the frequency of the tinnitus pitch.

threshold values are given in dB SPL, while the equal-loudness values and the ULLs (uncomfortable loudness levels) are given in dB SL.

The equal loudness judgments at the tinnitus-pitch frequency ranged from .4 to 43.0 dB SL. For 13 of 16 subjects the loudness match was to a tone of less than 10 dB SL. The ULLs ranged from 5.2 to 66.6 dB SL. The equal loudness judgments at the frequency of the most normal pure-tone threshold ranged from -.6 to 62 dB SL. Presumably the negative value is due to measurement error and should be just above threshold. Only four subjects judged their tinnitus to be equal in loudness to a pure tone of less than 10 dB SL. The ULLs ranged from 8.4 to 89.8 dB SL.

Table 5 displays the results for the contralateral measurements. At the frequency of the tinnitus pitch the equal-loudness judgments ranged from 1.4 to 38.8 dB SL. Ten of 15 subjects judged their tinnitus to be equal in loudness to a pure tone of less than 10 dB SL. The ULLs ranged from 8.8 to 71.2 dB SL. The equal-loudness judgments at the frequency of the most normal pure-tone threshold ranged from -.2 to 53 dB SL. Only two of 16 subjects judged their tinnitus to be equal in loudness to a pure tone of less than 10 dB SL. The ULLs ranged from 6.4 to 84.6 dB SL.

The individual ULLs can be converted from SL to SPL by adding the appropriate threshold values. When this was done, we noted that many subjects had ULLs of less than 80 dB SPL, and in one instance a ULL of 46 dB SPL was reported (Subject 7; contralateral ear at the frequency of the most normal pure-tone threshold).

Three examples of individual data are shown in Figure 1. The left side of the top panel contains the results for the ipsilateral ear for Subject 15. First note the data at the frequency of the tinnitus pitch. The threshold is about 79 dB SPL, the equal-loudness match 1 dB SL,

and the ULL 20 dB SL. Now examine the data at the frequency of the most normal pure-tone threshold. The threshold is about 20 dB SPL, the equal-loudness match is 50 dB SL, and the ULL is 90 dB SL. A larger dB SL measurement for the equal-loudness judgment is found at the most normal frequency than is found at the pitch-match frequency. This is also observed in the upper right panel, where the data for the contralateral ear are shown.

The middle panel of Figure 1 shows the data for Subject 5. Again, the tinnitus loudness match is at a very low dB SL at the frequency of the tinnitus pitch but at a much greater dB SL at the frequency of the most normal pure-tone threshold. This trend could be observed in 11 of our 16 subjects. The bottom panel displays the results for Subject 2, who is representative of four of our subjects (2, 12, 13, and 14). On the left side the loudness match is 3 dB SL at the tinnitus frequency and 4 dB SL at the frequency of the most normal pure-tone threshold. In this case the loudness match is at a low dB SL, even where pure-tone thresholds are normal.

Masking

Figure 1 also displays the minimum level of broadband noise required to mask the tinnitus. Subject 15 (upper panel) required a high-level masker, whereas Subject 5 (middle panel) required only a low-level masker, despite similarities in their audiograms and in their most normal frequencies. Subject 2 (bottom panel) required only a low-level masker, because of a moderate pure-tone threshold loss at 8000 Hz. In all three subjects the noise level required to mask the tinnitus was about the same in both the ipsilateral and contralateral ears.

Table 6 displays the individual results for broadband noise stimuli. The broadband noise thresholds range

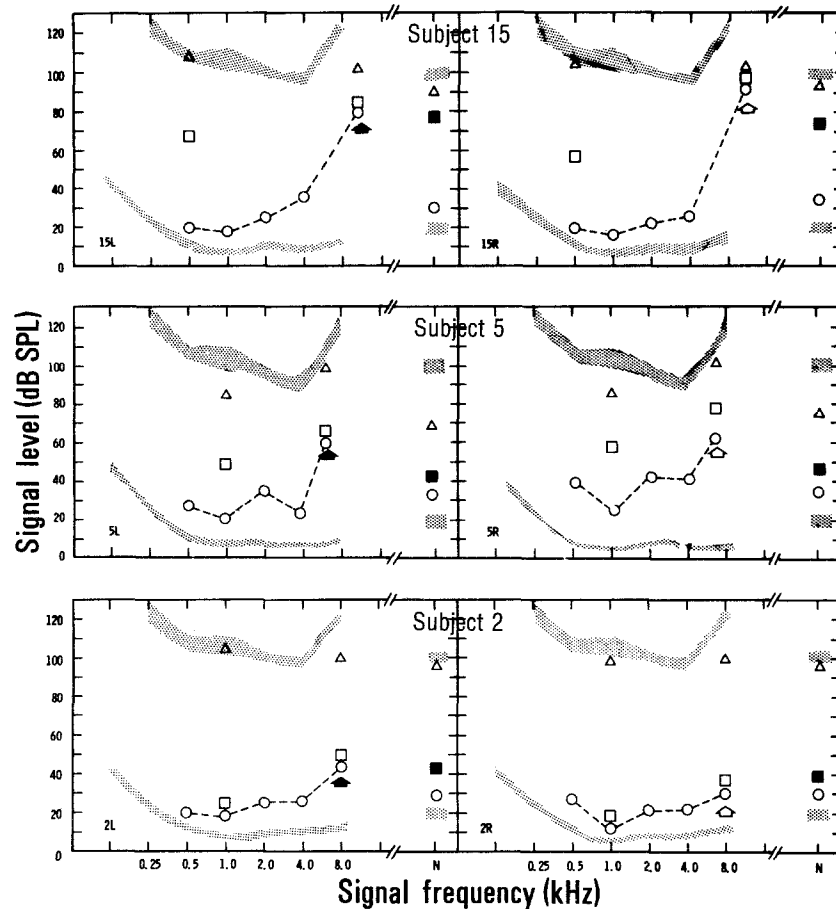


FIGURE 1. In each panel, the thin shaded line along the bottom represents normal pure-tone thresholds, and the shaded area at the top represents the ULL for normals. Also shown are subjects' threshold (O), ULL (Δ), and level judged equally as loud as the tinnitus ($\square$). The right of each panel shows the results for the broadband noise (N). The frequency of the pitch match is indicated by $\blacktriangle$ for the ipsilateral ear, and $\triangle$ for the contralateral ear (L = left, R = right).

TABLE 6. Tinnitus broad-band noise masking results (in dB SL) obtained from 14 subjects.

Subject	Ipsilateral			Contralateral		
	Threshold	Masked	ULL	Threshold	Masked	ULL
2	23.3	10.4	71.7	22.7	5.3	95.0
3	32.3	40.4	48.7	27.0	50.0	57.7
5	35.3	11.0	34.7	35.3	11.7	41.7
6	40.3	12.4	25.4	44.0	5.7	18.0
7	34.0	10.3	15.7	29.7	17.3	20.6
8	31.7	16.6	28.6	23.0	24.3	39.3
9	35.5	26.0	40.7	20.2	23.0	66.3
10	22.7	33.0	72.3	22.7	38.0	72.3
11	50.0	14.0	22.3	51.0	5.0	18.7
12 ^a	38.3	1.0	12.0	—	—	—
13	55.7	3.0	10.3	72.5	.3	2.3
14	39.3	10.4	11.7	38.3	12.0	12.7
15	21.6	48.7	62.1	24.7	43.7	60.3
16	55.0	17.7	29.7	55.0	4.0	28.3
$\bar{x}$	36.8	18.2	34.7	35.9	18.5	41.1
SD	10.9	13.9	21.6	15.8	16.7	27.4

Note. Masking data were not obtained for Subjects 1 and 4 due to time constraints.

^aSubject 12 was unable to hear the noise in the contralateral ear because of a hearing loss.

from 20.2 to 72.5 dB SPL and are correlated with the most normal pure-tone threshold in the same ear ($r = .932, p < .001$ for the contralateral ear; $r = .800, p < .001$ for the ipsilateral ear). This compares favorably with the findings of Hinchcliffe (1961), who found a correlation of around $r = .9$ between thresholds for 1000-Hz tones and those for broadband noise in normal listeners.

The minimum noise level required to mask the tinnitus ranged from 1.0 to 48.7 dB SL in the ipsilateral ear and from .3 to 50.0 dB SL in the contralateral ear. The effectiveness of the noise in the contralateral ear is remarkable, especially for the eight subjects whose tinnitus was localized to one ear or one side of the head. Figure 2 displays the relationship between the noise levels required to mask tinnitus in the ipsilateral and contralateral ears. In many subjects the minimum noise level in dB SL required to mask the tinnitus was about the same in both ears, regardless of which ear contained the tinnitus. The correlation coefficient between these two measures is $r = .906 (p < .001)$. Subject 16, who had a bilateral tinnitus that was louder in the left ear, required a 17.7-dB SL (72.7-dB SPL) masker in the ipsilateral ear but only a 4.0-dB SL (59.0-dB SPL) masker in the contralateral ear.

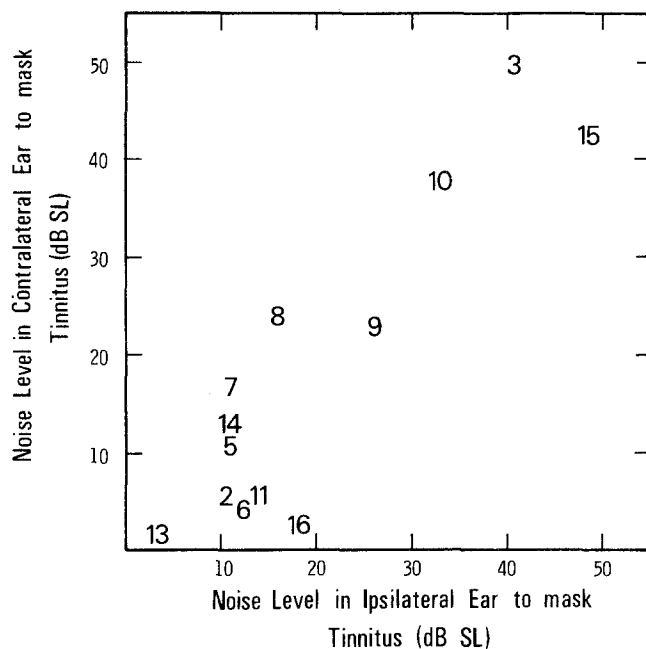


FIGURE 2. The level (in dB SL) of broadband noise required to just mask the tinnitus in the ear ipsilateral and contralateral to the tinnitus.

DISCUSSION

Our results indicate that the SL of a tone matched in loudness to the tinnitus depends greatly on the pure-tone frequency. Similar findings have been reported by Goodwin and Johnson (Note 2). Because of these inconsistencies, and because SL measurements cannot be in-

terpreted as loudness, we suggest adopting a sone scale for the loudness of tinnitus. What is required is an equation that can be used to convert the results of a loudness balance procedure (in dB SL) into the loudness scale (in sones). First, we describe the normal growth in loudness of a 1000-Hz tone. This defines the relationship between dB SPL and loudness. Second, we describe the relationship between dB HL and loudness for normal listeners. Here we include frequencies other than 1000 Hz. Third, we discuss different possible forms of the loudness functions observed in cochlear pathology and indicate how these can be applied to tinnitus loudness. Fourth, we then use some of these equations to transform the dB SL measurements from our study to tinnitus loudness in sones. Finally, we discuss the broadband noise masking of tinnitus and its relationship to tinnitus loudness.

Loudness and dB SPL

The *loudness level* of a sound (in phons) is equal to the SPL of a 1000-Hz tone judged to be equally loud. Thus, if a 500-Hz tone is judged to be as loud as a 40-dB SPL 1000-Hz tone, then the loudness level of the 500-Hz tone is 40 phons. Because the standard frequency is 1000 Hz, the loudness level (in phons) of a 1000-Hz tone is equal to its SPL. The solid line in Figure 3 (left) displays the relationship between the *loudness level* (in phons) and *loudness* (in sones). The sone scale is a ratio scale, with a sound having a loudness of 2 sones being twice as loud as a sound of 1 sone. This function is based on several studies using halving and doubling procedures and magnitude estimates (for a review, see Stevens, 1955) and is accepted as an international standard (ISO R131-1959). Note that each 10-dB SPL increase of a 1000-Hz tone doubles its loudness. The equation for this solid line is:

$$L = kP^{.6},$$

where L is the loudness in sones, k is a constant that depends on the units, and P is the sound pressure. With P measured in micropascals, k is .01. This formula indicates a simple power law relationship between loudness and sound pressure (i.e., the loudness increases with the sound pressure by a power of .6). This figure displays the loudness function for a 1000-Hz pure tone. If the loudness function of a tone of some other frequency is desired, the other tone must first be matched in loudness to a 1000-Hz tone.

Although there is some disagreement over the exact shape of the loudness function at low SPLs (Lochner & Burger, 1961; Marks & Stevens, 1968; Scharf & Stevens, 1959), it is agreed that loudness grows more rapidly near threshold than is predicted by Stevens's power law. The dashed line in Figure 3 (left) displays one model of loudness growth near threshold (Scharf & Stevens, 1959) and is described by the equation:

$$L = k (P - P_0)^{.6} \quad (1)$$

P_0 is meant to bring the zero of the physical scale into

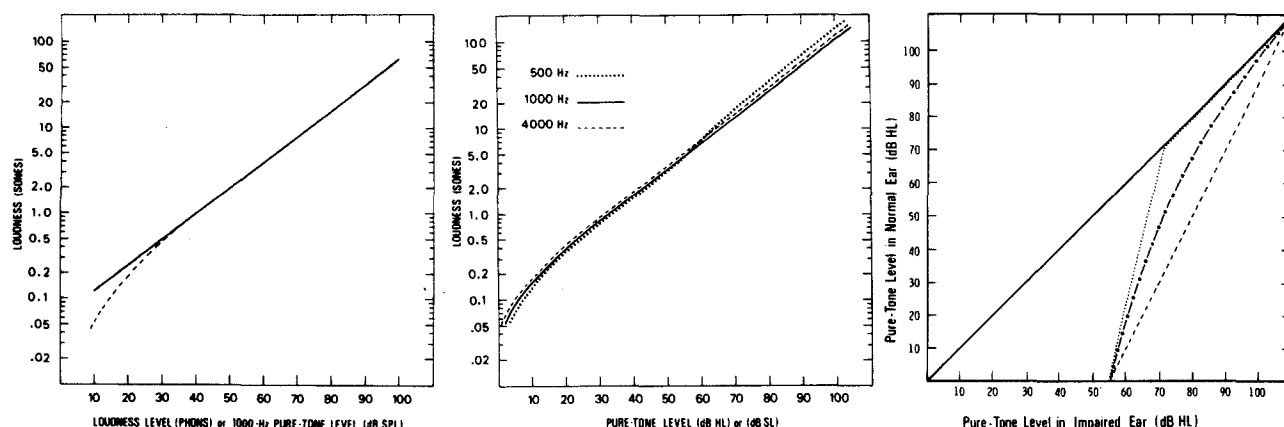


FIGURE 3. Left: The relationship between loudness in sones and loudness level (in phons) for a 1000-Hz pure tone. The loudness level is equal to the SPL of a 1000-Hz tone; therefore, the abscissa can also be labelled in dB SPL. The solid line represents the equation $L = kP^6$, where L is the loudness, k is a constant that depends on the units, and P is the pressure. The deviation at levels close to threshold, shown by the dashed line, indicates a more rapid growth of loudness near threshold and is a better approximation to experimental data. The equation for the dashed line is $L = k(P - P_0)^6$ (from Scharf & Stevens, 1959) where P_0 is subtracted from P to achieve the rapid loudness growth threshold. Center: The relationship between loudness (in sones) and dB HL for a 1000-Hz tone (solid line), a 500-Hz tone (dashed line), and a 4000-Hz (dotted line). For a normal ear with a pure-tone threshold of 0 dB HL, the abscissa can also read in dB SL. Right: Hypothetical examples of loudness recruitment as measured by the Alternate Binaural Loudness Balance procedure. The level of tone in the normal ear (shown along the ordinate) is equated in loudness to a tone in the impaired ear (whose level is shown along the abscissa). The solid line displays the expected results for someone with normal hearing in both ears. The three broken lines display different possible forms of loudness recruitment for individuals with one normal ear and one hearing-impaired ear.

line with the zero of the subjective scale. We use this relationship to describe loudness growth in terms of dB HL and dB SL (for normal listeners) in the next section.

Loudness and dB HL in Normal Ears

We can easily convert the dB SPL plotted along the abscissa in Figure 3 (left) to dB HL using the reference equivalent SPLs provided in ISO standard 389-1975. For a Telephonics TDH-49 earphone encased in an MX41/AR cushion, this is accomplished by subtracting 7.5 dB. The solid line (center) represents this modification and displays the loudness of a 1000-Hz tone as a function of dB HL.

It is also possible to include the growth of loudness for other frequencies on this graph using published data on equal-loudness contours (Stevens & Davis, 1938). The results of these conversions show a similar growth in loudness for pure tones between 500 Hz (dotted line) and 4000 Hz (dashed line). This is generally true for all frequencies between 500 and 4000 Hz. For frequencies less than 500 Hz or greater than 4000 Hz, however, the dynamic range of the auditory system decreases, and the loudness functions diverge from the 1000-Hz loudness function.

This graph assumes a normal loudness function, but it is known (e.g., Hallpike & Hook, 1951; Harris, Haines, & Myers, 1952; Priede & Coles, 1974) that cochlear-impaired listeners exhibit abnormal loudness functions. We discuss how the loudness function can be modified by pathology in the next section.

Possible Forms of Abnormal Loudness Functions

The abnormally steep loudness functions accompanying elevated pure-tone thresholds have long been taken as a clinical index of cochlear pathology. This phenomenon, referred to as "recruitment," has received a great deal of attention in the clinical literature (for a review see Carver, 1978; Harris, 1953). However, relatively little attention has been given to quantifying exact forms of abnormal loudness functions. In this section we describe some possible forms of loudness recruitment.

Figure 3 (right) displays examples of recruitment as measured clinically with the *Alternate Binaural Loudness Balance* (ABLB) procedure (Fowler, 1936; Hood, 1969; Jerger, 1962). The graph displays the levels of tones presented to the impaired and normal ears which are judged to be equal in loudness. Four ABLB functions are depicted. The solid diagonal line represents the results expected from an individual with normal ears. Equal levels in both ears are required to produce equal loudness. The other three lines represent possible ABLB functions obtained from individuals with a pure-tone threshold of 55 dB HL in one ear and normal hearing in the other. The dashed line displays a linear (on the log-log scales) relationship between the threshold and 110 dB HL. An ABLB function of this type was suggested by Harris et al. (1952) and Ward, Fleer, and Glorig (1961) for individuals with a noise-induced hearing loss.

Another possible form of recruitment is depicted by the dotted line. At low SLs the function is steeper than the normal loudness function. At high SLs the loudness growth is normal. Hallpike and Hood (1951) observed a

similar ABLB function in their data on listeners with Ménière's syndrome. They reported that the greater the pure-tone threshold loss, the greater the slope of the abnormal part of the ABLB function, in agreement with some previous observations of Reger (1935, 1936). Stevens and Guirao (1967) suggested further that the abnormal loudness growth was limited to within 30 dB of threshold, independently of the amount of pure-tone threshold loss.

The dashed-dot line represents yet a third possible abnormal ABLB function. Steinberg (1935), Steinberg and Gardner (1937), and Priede and Coles (1974) all suggested that a curvilinear approximation was a good description of the ABLB function of hearing-impaired individuals.

Even among individuals with identical pathologies and identical hearing levels there is no one typical or standard form of recruitment. Furthermore, seldom are enough data points collected in clinical practice to warrant a precise decision about which of these three types of functions would provide the best fit. Therefore, any single formula that describes the loudness function in hearing-impaired listeners must represent only a general approximation which may not be completely accurate for a specific individual. Even among normal listeners, Stevens and Tulving (1957) have shown large individual differences in the loudness function.

ABLB measurements do not actually depict loudness functions per se; they only show levels of tones judged to be equal in loudness. In cases with recruitment, we are able to deduce that the loudness function in the impaired ear is steeper than it is in the normal ear. We are not able to determine the actual loudness of a tone at any

level. What we require is a function relating dB HL to loudness (in sones) for an impaired ear, such as the one shown in Figure 3 (center) for normal ears.

Recruitment

One approach to this problem is to reexamine the loudness function in the normal ear, as described by equation 1. We noted the P_0 was introduced into Stevens's power law to equate the lowest audible level with 0 sones. This can be taken one step further for hearing impairment by letting P_0 equal the dB HL (in Pa) at threshold (see also Scharf, 1978; Steinberg & Gardner, 1937). Some examples of this approach are shown in Figure 4 (left) for hypothetical subjects with pure-tone thresholds of 0, 35, and 70 dB HL. The functions describe the estimated loudness of a tone at any level above threshold for individuals with specific pure-tone threshold losses. For example, a tone of 10 dB SL for the individual with a 35-dB HL threshold has an estimated loudness of nearly 2 sones. However, a tone of 10 dB SL for the individual with a 70-dB HL threshold has an estimated loudness of nearly 20 sones. In other words, a tone of 10 dB SL presented to the individual with a 70-dB HL threshold is estimated to be about 10 times louder than the same 10-dB SL tone presented to an individual with a 35-dB HL threshold.

There is one further point to make about the format of Figure 4. The top half of the right ordinate is labeled in dB HL. We are able to do this because for pure tones above about 30 dB SPL there is a linear relationship between the loudness (in sones) and dB SPL (see Figure 1). Therefore, the top half of Figure 4 can be compared

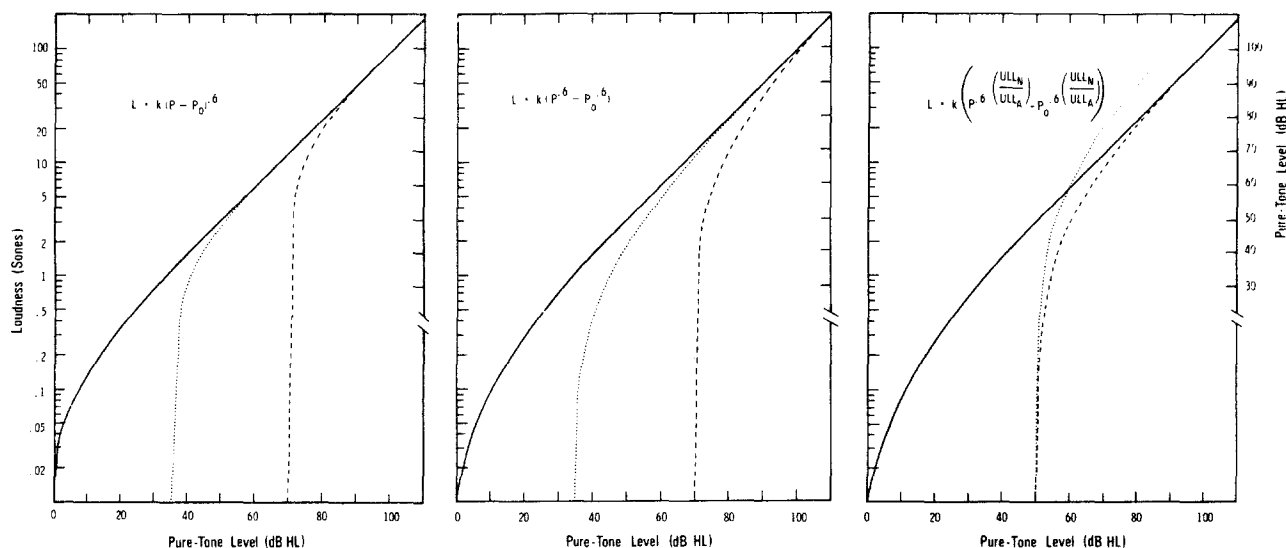


FIGURE 4. Left: Loudness growth according to the equation $L = k(P - P_0)^6$ (see text). The three functions represent an individual with pure-tone thresholds of 0 dB HL (solid line), 35 dB HL (dotted line), and 70 dB HL (dashed line). Center: Loudness growth according to the equation $L = k(P^6 - P_0^6)$ (see text). The three lines represent an individual with pure-tone thresholds of 0 dB HL (solid line), 35 dB HL (dotted line), and 70 dB HL (dashed line). Right: Loudness growth according to the equation $L = k[P^6 (ULL_N/ULL_A) - P_0^6 (ULL_N/ULL_A)]$ (see text). The three lines represent examples of individuals with pure-tone thresholds of 0 dB HL, $ULL = 110$ dB HL (solid line); threshold = 50 dB HL, $ULL = 90$ dB HL (dotted line); and threshold = 50 dB HL, $ULL = 110$ dB HL (dashed line).

directly to the ABLB graphs shown in Figure 3 (right). The agreement between the ABLB functions predicted by this equation for levels greater than 30 dB HL and the ABLB functions shown in Figure 3 (right) is reasonable.

Several authors have commented on the similarities between recruitment and the loudness function in normal listeners in the presence of masking noise (Stevens, 1966; Stevens & Guirao, 1967). Although differences may exist (e.g., Richards, 1973), the fact that there is no single function defining recruitment in all hearing-impaired listeners makes the analogy worth pursuing. We have, therefore, considered these experiments in our attempt to define the abnormal loudness function. Lochner and Burger (1961) suggested the formula:

$$L = k (P_o^6 - P_o^6) \quad (2)$$

to describe the loudness function in quiet and in the presence of broadband noise. P_o is simply the masked threshold.

Figure 4 (center) shows our application of equation 2 for hypothetical individuals with pure-tone thresholds of 0 dB HL, 35 dB HL, and 70 dB HL. These loudness functions are slightly less steep than those shown in Figure 4 (left). The loudness of a tone at 10 dB SL for the listener with a pure-tone threshold of 35 dB HL is estimated to be about 1 sone, while for the listener with a pure-tone threshold of 70 dB HL, a tone of 10 dB SL is estimated to be about 12 sones.

Other formulae to describe the loudness function in normal ears (with and without masking) have been suggested by Zwislacki (1965), Hellman and Hellman (1975), and Macrae (1979). These formulae are generally more complex, and although each has some theoretical merit, we chose to use relatively simple models for our initial attempt to describe the abnormal loudness function.

Loudness Growth and Reduced ULLs

We suggest one further consideration in describing the loudness function in hearing-impaired listeners. Some individuals display ABLB functions in which the level of the tone presented to the impaired ear is less than the level presented to the normal ear to achieve a loudness balance. This has been called "over-recruitment" (e.g. Day, 1950; Denes & Naunton, 1959; Eby & Williams, 1951) and may indicate that the impaired ear reaches its ULL at a lower dB HL than the normal ear.

Several of our tinnitus subjects displayed very low ULLs (uncomfortable loudness levels). All subjects were given identical instructions, and all values represent averages of five replications. We considered three possible explanations for these reduced ULLs. The first is that the instructions may have been misunderstood. During testing the instructions were read to each subject exactly as they appear in the Method section. These instructions were repeated if the subject experienced difficulty in making judgments, but for some inexperienced, older subjects this may have been insufficient. Secondly, the

lesion causing tinnitus could have some direct effect on the loudness perception that leads to a reduced ULL. Thirdly, having to listen to tinnitus for many years may have rendered some individuals more annoyed by continuous, unnatural sounds in general. If either of the last two possibilities is correct, then perhaps the reduced ULL and hence the reduced dynamic range should be considered in describing the loudness of tinnitus. Poulton (1967) and Teghtsoonian (1971) suggested that the dynamic range of a sensory modality influences the slope of the sensation-magnitude function.

If we consider the ULL as the upper limit of the loudness function (and assume that all ULLs are equally loud at about 200 sones), then we can examine the loudness growth functions for two hypothetical individuals with pure-tone thresholds of 50 dB HL but with two different ULLs, one of 90 dB HL and one of 110 dB HL. We have modified the exponents in equations 1 and 2 to obtain:

$$L = k (P - P_o)^{6x} \quad (3)$$

$$L = k (P^{6x} - P_o^{6x}) \quad (4)$$

where x equals the normal ULL (ULL_N) in dB HL, divided by the abnormal ULL (ULL_A).

In the example shown in Figure 4 (right), which uses equation 4, we have let $ULL_N = 110$ dB HL. A tone of 10 dB SL for the hypothetical subject with a ULL of 110 dB HL is about 3 sones. A tone of 10 dB SL for the subject with a ULL of 90 dB HL and the same threshold is estimated to be about 6 sones. The formula clearly provides that with a reduced dynamic range (i.e., lower ULL), there will be a steeper loudness function than there is in subjects with a normal ULL.

Tinnitus Loudness in Sones

We restricted the tone frequency used for the most normal pure-tone threshold to between 500 and 4000 Hz because the loudness growth in normal listeners in quiet and under masking can be closely approximated by equations 1 or 2. However, many of the pitch-match frequencies were beyond this range, and thus these formulae may be inappropriate for these cases. Nevertheless, it might be that P_o , a correction for the threshold elevation and the reduced dynamic range, allows for a reasonable approximation to the *abnormal* loudness function independent of signal frequency. Without sufficient data from hearing-impaired listeners to define the loudness growth precisely, and in the interest of simplicity, we have applied the formulae to frequencies outside the 500-4000-Hz range.

Table 7 shows the conversion of the equal loudness matches in dB SL taken at the tinnitus frequency in the ipsilateral ear into the tinnitus loudness (in sones) using the four formulae that we have proposed. Subject 1 had the highest loudness match at 43.0 dB SL, and Subject 13 had the lowest loudness match of .4 dB SL. Thirteen of the values are less than 10 dB SL. Equation 1 redistributes the order of tinnitus loudness, with Subject 12 having the loudest tinnitus, 75 sones, despite a loudness

TABLE 7. The dB SL judged equally as loud as the tinnitus and the tinnitus loudness in sonos determined for the ear ipsilateral to the tinnitus at the tinnitus-pitch frequency. See text for explanation of the four equations used to derive the loudness in sonos. $x = ULL_N/ULL_A$.

Subject	Sensation level (dB SL)	$L = k (P-P_o)^{.6}$ (sones)	$L = k (P^{.6} \cdot P_o^{.6})$ (sones)	$L = k (P-P_o)^{.6x}$ (sones)	$L = k (P^{.6x} \cdot P_o^{.6x})$ (sones)
1	43.0	3.86	4.63	40.27	39.75
2	.6	.23	.05	.38	.12
3	8.4	17.22	10.10	43.00	28.42
4	9.6	3.58	2.21	9.34	6.64
5	7.0	4.70	2.57	13.39	8.76
6	1.8	8.28	2.65	47.74	24.61
7	4.4	2.81	1.28	145.66	150.73
8	12.4	26.74	18.14	159.63	127.01
9	7.4	.79	.44	382.46	607.81
10	5.2	12.91	6.28	44.36	26.71
11	7.6	.98	.55	10.01	8.92
12	5.2	75.65	36.88	95.69	48.22
13	.4	3.24	.57	19.30	7.83
14	33.6	6.81	6.22	115.40	113.30
15	1.2	4.75	1.29	12.62	4.80
16	1.0	5.58	1.41	37.19	18.03
$\bar{x}$	9.3	11.20	5.96	73.53	76.35
SD	11.97	18.51	9.40	93.24	149.33

match of only 5.2 dB SL. This tinnitus loudness is equivalent to the loudness of a 1000-Hz pure tone of over 100 dB SPL in a normal ear. Subject 2 had the "softest" tinnitus, .23 sonos, with a loudness match of .6 dB SL. This tinnitus loudness is equivalent to the loudness of a 1000-Hz tone of about 32 dB SPL in a normal ear.

The application of equation 2 results in slightly lower loudness estimates than does equation 1. The application of equations 3 and 4 results in loudness estimates that are greater than those predicted by equations 1 and 2, particularly in those individuals with very low ULLs.

We have no direct measure of the tinnitus loudness with which to compare the results of our conversions to sonos. One way of evaluating the validity of our formulae would be to examine the consistency of loudness values determined at two different frequencies in each ear for the individual results. These four measurements, although having different dB SL values, should all be of equal loudness and therefore be converted to similar sonos values. In several subjects this was indeed the case. For example, Subject 3 had equal loudness matches of 8.4, 32.0, 6.2, and 24.2 dB SL at the two tinnitus frequencies and the two most normal frequencies (ipsilateral and contralateral values, respectively). When these values are converted to sonos (for example, with equation 2), the tinnitus loudness estimates at the two frequencies in the two ears convert to 10.1, 6.8, 7.3, and 3.1 sonos. Similarly, Subject 15's measurements of 1.2, 44.8, 1.8, and 36.8 dB SL convert to 1.29, 6.4, 4.66, and 6.41 sonos.

Unfortunately, our conversions of the four measurements were in poor agreement for other subjects. For example, Subject 8's measurements of 12.4, 23.6, 5.81, and 28.4 dB SL converted to 18.14, 6.02, 5.75, and 2.35 sonos. There are several possible reasons for these inconsistencies. One contributing factor is that although

the pure tones at the different frequencies in the different ears were all matched with the loudness of the tinnitus, they were not balanced in loudness to each other. The assumption that they are equally loud may be invalid. We expect some imprecision in subjects' ability to match their tinnitus loudness. Even errors of 2-3 dB would result in considerable differences in sone values at low dB SL values. A few of our subjects demonstrated low dB SL matches irrespective of their pure-tone thresholds. This suggests that in some subjects the effects of loudness recruitment may not explain adequately the low dB SL matches of tinnitus loudness. These subjects may have a different category of tinnitus, for which loudness matching is an inappropriate technique to determine tinnitus loudness. Finally, the formula chosen to represent an *average* recruitment function may be too general. The equation to describe the loudness function may vary across frequency in the same subject and among hearing-impaired listeners. We have assumed that all the subjects display a similar loudness function. It may be that varying degrees of partial- or over-recruitment were present. It may also be that equations 1 and 2 cannot be used at pitch-match frequencies beyond the 500-4000-Hz range. By using a single formula that estimates an average form of recruitment, we may be able to determine only the loudness of tinnitus within a range of from one-half to twice our calculated sone value. Nevertheless, we feel that this at least provides us with the appropriate framework from which to view tinnitus loudness.

Loudness Category Scaling

Subjects rated their tinnitus loudness on a scale of 1-6, 1 being very soft and 6 being very loud. These category

scaling techniques are known to be highly nonlinear and should not be expected to produce equal interval or equal ratio scales (Stevens, 1975). In Figure 5 we display these loudness ratings as a function of the level of a tone judged to be equally as loud as the tinnitus in dB SL (for the results at the tinnitus frequency in the ipsilateral ear). Thirteen subjects matched their tinnitus loudness to a pure tone of less than 10 dB SL, despite the wide range of loudness estimates.

Figure 6 shows the loudness ratings as a function of the tinnitus loudness match in sones (using equation 2). Note the wide range of tinnitus loudness estimates in sones compared to the limited range in dB SL shown in Figure 5.

There are at least three important advantages to measuring tinnitus loudness in sones. First, these sone values have a communicable meaning. For example, Subject 4 had a tinnitus loudness of 2 sones. People with normal hearing can listen to a 1000-Hz pure tone at 50 dB SPL and experience the same loudness as the tinnitus experienced by Subject 4. Second, the sone values are more meaningful for expressing relative loudness than are dB SL measurements. For example, Subjects 9 and 10 in Figure 5 have both matched their tinnitus to a tone of about 9 dB SL. Because of the elevated pure-tone threshold of Subject 10, however, the tinnitus loudness in sones is estimated to be about 40 times louder for Subject 10 than for Subject 9 (see Figure 6). Third, some subjects reported that their tinnitus was extremely loud even though matching measurements suggested it to be quiet compared to the tinnitus experienced by others. Subject 13, for example (see Figure 6), indicated that his tinnitus has a loudness rating of 6, although his derived tinnitus loudness (in sones) was quieter than the estimated sone values of the tinnitus loudness experienced by three quarters of our subjects. The contrast between these two forms of measurement may highlight individuals for whom counselling needs to be emphasized.

Contralateral Masking of Tinnitus

It is remarkable that the broadband masking noise masked the tinnitus so effectively when the masker was presented to the contralateral ear. For subjects who reported bilateral tinnitus or tinnitus "in the head," the source of tinnitus was likely to be retrocochlear; therefore, we would expect masking to be effective in either ear. However, for subjects who localized tinnitus to one ear or the other, the reason for the effectiveness of the contralateral masking is less clear.

In normal-hearing subjects without tinnitus, a pure tone presented to one ear can be masked by a broadband noise presented to the contralateral ear in at least two different ways. First, the masker could be of sufficient energy to vibrate the skull and stimulate the opposite ear by bone conduction (e.g., Dirks, 1973; Tonndorf, 1972). For broadband noise, the noise level must be at least 40 dB SPL for this "cross hearing" to occur (Liden, Nilsson, & Anderson, 1959) and hence for cross-masking to begin

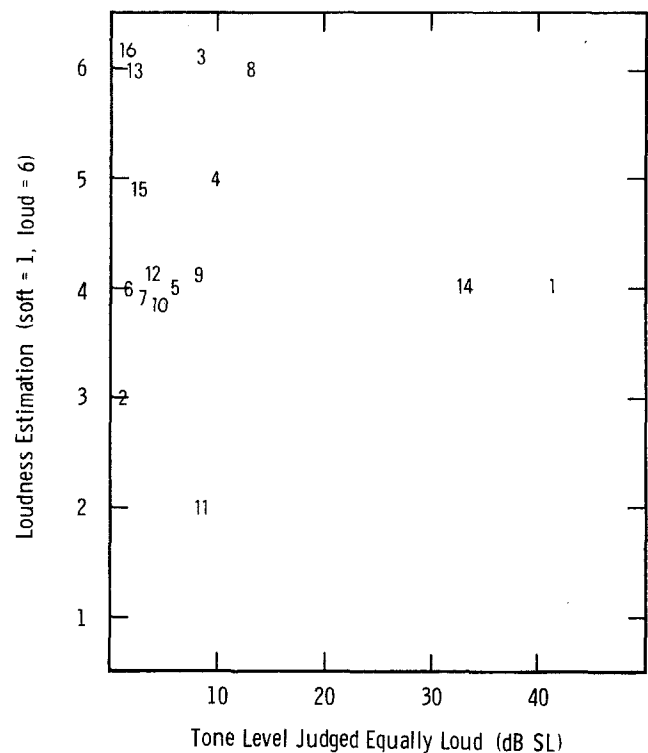


FIGURE 5. Loudness estimates as a function of the tone level judged equally as loud as the tinnitus (in dB SL) performed at the tinnitus frequency in the ear ipsilateral to the tinnitus.

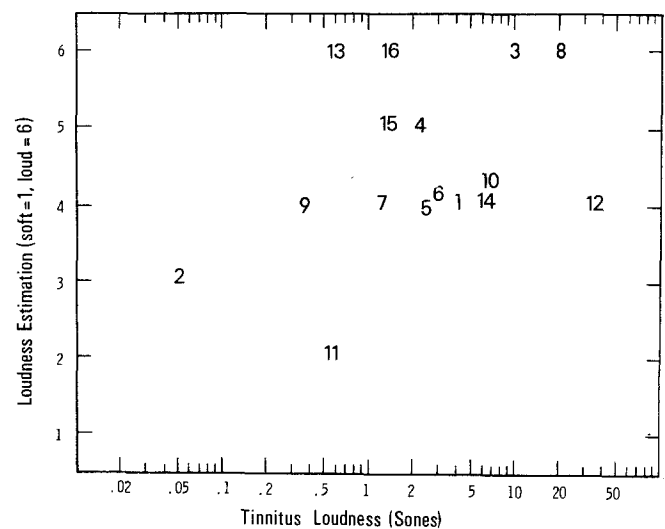


FIGURE 6. Loudness estimates as a function of the tinnitus loudness (in sones) derived from measurements at the tinnitus frequency in the ear ipsilateral to the tinnitus. Equation 2, $L = k (P_o^{0.6})$, was used to convert dB SL measurements into sones.

to operate. In the present experiment, the noise level in the contralateral ear would have to equal at least the sum of two quantities: the 40 dB SPL (required to cross over to the tinnitus ear) plus the noise level (in dB SL) required to mask the tinnitus in the ipsilateral ear. This situation never occurred in our study; thus, some other cause must be identified for the effectiveness of the contralateral masking.

The second way in which a noise presented to one ear can mask a signal presented to the other ear is by some form of neural interaction at the level of, or central to, the cochlear nucleus. Central masking (e.g., Dirks & Malmquist, 1965; Hughes, 1940; Zwislocki, 1953) of one tone on another is known to be highly frequency dependent (e.g., Zwislocki, 1970), but it has been reported that contralateral masking of tinnitus is not (Feldmann, 1971). This central-masking effect is usually rather small: For example, a 70-dB SPL noise produces only an 8-14-dB shift in threshold (Zwislocki, 1970). However, it could be an important source of the contralateral masking of a sensorineural tinnitus of peripheral origin.

It may also be that tinnitus localized in one ear has a retrocochlear origin, which could explain the effectiveness of the contralateral masking. This would be consistent with three observations: (a) that tinnitus localized to one ear can remain unchanged after auditory-nerve section (House & Brackmann, 1981); (b) our clinical observation that the masking of tinnitus localized to one ear sometimes results in the appearance of tinnitus in the other ear; and (c) that tinnitus localized to one ear can display different masking characteristics than tone-on-tone masking (Tyler, 1982).

The Relationship Between Masking and Tinnitus Loudness

Generally, the greater the intensity of a pure tone, the greater its loudness and the more intense the broadband noise required to mask the tone. It could be informative as well as clinically useful if a relationship could be established between tinnitus loudness and the minimum noise level required to mask the tinnitus. However, we found no correlation between the level of the pure tone (in dB HL) judged to be equal in loudness to the tinnitus and the minimum level of the noise (in dB SPL) required to mask the tinnitus. A relationship was found between these measurements when they were expressed in dB SL. In Table 8 we display correlation coefficients between the minimum noise level required to mask the tinnitus and the tinnitus loudness in dB SL and in calculated sones.

The dB SL matches obtained at the most normal fre-

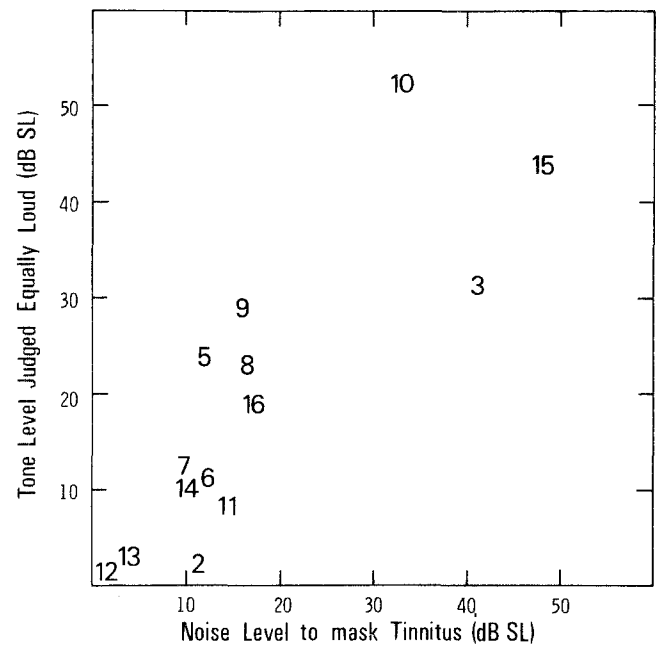


FIGURE 7. The noise level in the ipsilateral ear required to just mask the tinnitus as a function of the pure-tone level (at the frequency with the most normal pure-tone threshold) judged to be equal in loudness to the tinnitus (in dB SL). The correlation coefficient for the data shown is $r = .83$ ($p < .01$).

quency correlate significantly with the minimum noise level (in dB SL) required to mask the tinnitus. This relationship, further depicted in Figure 7, suggests that both measures may be related to the magnitude of the tinnitus. We avoid the term *loudness* at this point because we have argued that dB SL cannot be interpreted as loudness. The dB SL measurements at the tinnitus frequency do not correlate with the noise level required to mask the tinnitus, supporting our contention that this measure is influenced heavily by the pure-tone threshold loss. It may be that the noise level (in dB SL) required to mask the tinnitus can be converted to an equivalent tinnitus loudness with the use of a recruitment formula. However, we have not attempted this ourselves since the loudness function of broadband noise in the hearing-impaired ear will be complex.

The tinnitus loudness estimates in sones derived from

TABLE 8. Correlation coefficients between the noise level (in dB SL) required to just mask the tinnitus and the estimates of tinnitus loudness. Shown are the coefficients for the ipsilateral measurements and the coefficients for the contralateral measurements. $x = ULL_N/ULL_A$.

Frequency	dB SL	$L = k(P-P_o)^{.6}$	$L = k(P^{.6}-P_o^{.6})$	$L = k(P-P_o)^{.6x}$	$L = k(P^{.6x}-P_o^{.6x})$
Ipsilateral					
most normal	.83**	.45	.66*	.06	.06
tinnitus pitch	-.08	.35	.26	.18	.19
Contralateral					
most normal	.72**	.59*	.62*	-.14	-.12
tinnitus pitch	.07	.61*	.46	.09	.08

* $p < .05$.

** $p < .01$.

formulae 1 and 2 correlate moderately with the masking level for measurements in both the ipsilateral and contralateral ear. This suggests only a weak relationship between our derivation of tinnitus loudness and the minimum noise level required to mask the tinnitus. It may be that either tinnitus masking and tinnitus loudness are unrelated, or that our formulae do not represent good approximations of tinnitus loudness.

The failure of formulae 3 and 4 to predict the masking levels suggests either that we have incorrectly modified the formula to account for abnormally low ULLs, that the abnormal ULLs have no direct bearing on the perception of tinnitus loudness, or that the low ULLs were not valid estimates of the upper limit of the dynamic range.

SUMMARY

We have argued that dB SL measurements are inappropriate estimates of tinnitus loudness because they fail to consider the loudness recruitment observed in hearing-impaired listeners. We have presented formulae that convert dB SL measurements, given the pure-tone threshold, into tinnitus loudness in sones. Additional formulae were presented which considered the reduced ULLs that we observed in some subjects.

These formulae have the advantage of representing tinnitus loudness on an absolute scale so that the tinnitus loudness of different tinnitus sufferers can be compared. Furthermore, the tinnitus loudness can be compared directly to the loudness of a 1000-Hz pure tone experienced by normal subjects. The minimum level of a broadband noise required to mask the tinnitus was found to be similar (in dB SL) in the ear ipsilateral and in the ear contralateral to the tinnitus. This suggests that some form of central masking effect is operative in tinnitus and that tinnitus localized in one ear might have a retrocochlear locus.

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